Mathematical Specification: Spatial CRT Design Simulation with Heterogeneous Incidence

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1 Introduction

This document provides a rigorous mathematical specification for the simulation study evaluating treatment assignment designs in Spatial Cluster Randomized Trials (CRTs) under heterogeneous outcome incidence. The study investigates which sampling design strategy yields the most efficient (lowest error) estimates of the direct intervention effect τ when spillover effects and spatial dependence are present.

Research question: Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment design minimizes estimation error for τ across a range of spatial dependence and spillover configurations?

Application context: Sudden Unexpected Death (SUD) in North Carolina, where county-level and census tract-level incidence rates are modeled via Poisson regression (Gan et al., 2019; Watson et al., AHA Abstract). The base SUD incidence rate is approximately 35 per 100,000 (Mirzaei et al., 2019).

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2 Study Region and Spatial Structure

2.1 Regular Lattice Grid

The study region is a regular square lattice grid:

$$\mathcal{G} = \{s_1, s_2, \ldots, s_N\}$$

where each cluster s_i is located at integer coordinates (x_i, y_i) on a $G \times G$ grid, so $N = G^2$. In this study, $G = 10$ and $N = 100$.

2.2 Spatial Contiguity

Two contiguity definitions are considered:

- **Rook contiguity**: clusters sharing an edge (4-connected). The neighbor set is

$$\mathcal{N}_R(s_i) = \{s_j : |x_i - x_j| + |y_i - y_j| = 1\}$$

- **Queen contiguity**: clusters sharing an edge or vertex (8-connected). The neighbor set is

$$\mathcal{N}_Q(s_i) = \{s_j : \max(|x_i - x_j|, |y_i - y_j|) = 1\}$$

2.3 Spatial Weight Matrix

Let $\mathbf{A}$ denote the binary adjacency matrix where $A_{ij} = \mathbf{1}[s_j \in \mathcal{N}(s_i)]$. The row-standardized spatial weight matrix is:

$$W_{ij} = \frac{A_{ij}}{\sum_{k=1}^{N} A_{ik}}, \quad i \neq j; \quad W_{ii} = 0$$

This ensures $\sum_j W_{ij} = 1$ for each row, so spatial lags represent neighborhood averages. We construct $\mathbf{W}_R$ (rook) and $\mathbf{W}_Q$ (queen) weight matrices.

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3 Outcome Data Generating Process

3.1 Spatial Durbin Model (SDM)

The outcome vector $\mathbf{Y} = (Y_1, \ldots, Y_N)^\top$ is generated from the Spatial Durbin Model:

$$\mathbf{Y} = (\mathbf{I}_N - \rho\mathbf{W})^{-1} \left[\tau\mathbf{Z} + \gamma \cdot \text{Spill}(\mathbf{Z}) + \beta\mathbf{X} + \boldsymbol{\varepsilon}\right]$$

where:

Symbol	Description	Role
$\mathbf{Y}$	$N \times 1$ outcome vector	Response
$\mathbf{I}_N$	$N \times N$ identity matrix	—
ρ	Spatial autocorrelation parameter	Controls global spatial dependence in $\mathbf{Y}$
$\mathbf{W}$	$N \times N$ row-standardized weight matrix	Spatial structure (rook or queen)
τ	Direct treatment effect	**Estimand of interest**
$\mathbf{Z}$	$N \times 1$ binary treatment vector	$Z_i \in \{0, 1\}$
γ	Spillover effect magnitude	Indirect effect via treated neighbors
$\text{Spill}(\mathbf{Z})$	$N \times 1$ spillover term	See Section 3.2
β	Effect of baseline incidence on outcome	Covariate coefficient
$\mathbf{X}$	$N \times 1$ baseline incidence vector	$X_i \in [0, 1]$
$\boldsymbol{\varepsilon}$	$N \times 1$ error vector	$\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$

The spatial multiplier $(\mathbf{I}_N - \rho\mathbf{W})^{-1}$ induces global spatial dependence: each cluster's outcome depends on the entire network through higher-order neighbors.

Stationarity condition: The model requires $|\rho| < 1/\lambda_{\max}(\mathbf{W})$. For row-standardized $\mathbf{W}$, $\lambda_{\max} = 1$, so we require $|\rho| < 1$.

3.2 Spillover Construction

The spillover term captures indirect treatment effects transmitted through the spatial network:

$$\text{Spill}_i = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|} = (\mathbf{WZ})_i$$

This is the proportion of cluster i's neighbors that are treated. Two spillover types are considered:

1. **Control-only spillover**: The spillover effect applies only to untreated clusters:

$$\text{Spill}_i^{(\text{ctrl})} = \gamma \cdot (\mathbf{WZ})_i \cdot (1 - Z_i)$$

2. **Bidirectional spillover**: The spillover effect applies to all clusters:

$$\text{Spill}_i^{(\text{both})} = \gamma \cdot (\mathbf{WZ})_i$$

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4 Incidence Generation Modes

The baseline incidence vector $\mathbf{X}$ represents prior observed outcome rates across clusters. We consider three generation modes with increasing structural complexity. All modes produce values in $[0, 1]$ to ensure comparability across modes and with the treatment assignment designs.

4.1 Mode 1: Independent and Identically Distributed (iid)

$$X_i \overset{\text{iid}}{\sim} \text{Uniform}(0,1), \quad i = 1, \ldots, N$$

This serves as the **naive baseline** with no spatial structure in incidence. It represents the case where historical outcome rates show no geographic clustering.

Properties: $\mathbb{E}[X_i] = 0.5$, $\text{Var}(X_i) = 1/12$, $\text{Cor}(X_i, X_j) = 0$ for $i \neq j$.

4.2 Mode 2: Spatially Correlated (SAR Filter)

Spatially correlated incidence is generated via the Simultaneous Autoregressive (SAR) filter:

$$\mathbf{X}^* = (\mathbf{I}_N - \rho_X \mathbf{W})^{-1}\mathbf{u}, \quad \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$$

where $\rho_X \in [0, 1)$ controls the degree of spatial correlation in incidence.

The raw values X_i^* are then transformed to $[0, 1]$ via the probability integral transform:

$$X_i = \Phi(X_i^*)$$

where $\Phi(\cdot)$ is the standard normal CDF.

Key property: When $\rho_X = 0$, $\mathbf{X}^* = \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$ and $X_i = \Phi(u_i) \sim \text{Uniform}(0, 1)$ by the probability integral transform, recovering Mode 1 marginally. When $\rho_X > 0$, the X_i remain marginally approximately uniform but exhibit positive spatial correlation: nearby clusters have similar incidence values.

Parameters: $\rho_X \in \{0.20, 0.50\}$ (incidence spatial correlation).

4.3 Mode 3: Poisson Count-Based

Motivated by the SUD application, this mode generates incidence from a Poisson count model with spatially correlated rates:

Step 1. Generate spatially correlated log-rates:

$$\log \boldsymbol{\lambda}^* = \mu_0 \cdot \mathbf{1} + (\mathbf{I}_N - \rho_X \mathbf{W})^{-1}\mathbf{u}, \quad \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$$

where $\mu_0 = \log(\text{base rate})$ is the log of the baseline incidence rate (e.g., $\mu_0 = \log(35/100{,}000)$ for SUD).

Step 2. Generate Poisson counts given cluster populations:

$$C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$$

where P_i is the population of cluster i.

Step 3. Compute observed rates:

$$R_i = C_i / P_i$$

Step 4. Rank-normalize to $[0, 1]$:

$$X_i = \frac{\text{rank}(R_i) - 0.5}{N}$$

This quantile-based transformation preserves the ordering of rates while mapping to a uniform marginal distribution on $[0, 1]$.

4.3.1 Population Modes

Two population specifications are available:

1. **Equal population:** $P_i = P_0$ for all i (default: $P_0 = 1{,}000$)
2. **Heterogeneous population:** $P_i \sim \text{LogNormal}(\log P_0, \sigma_P^2)$, where $\sigma_P = 0.5$ provides moderate population heterogeneity

Parameters: $\rho_X \in \{0.20, 0.50\}$, base rate $= 35/100{,}000$, $P_0 = 1{,}000$, population mode $\in \{\text{equal}, \text{heterogeneous}\}$.

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5 Treatment Assignment Designs

Eight designs are evaluated. Each design produces a binary treatment vector $\mathbf{Z} \in \{0,1\}^N$. Designs 1–2 are deterministic given $\mathbf{X}$; Designs 3–8 involve randomization.

5.1 Design 1: Checkerboard (Strict Spatial Stratification)

$$Z_i = (x_i + y_i) \mod 2$$

Treatment is assigned in an alternating checkerboard pattern. Every treated cluster is surrounded entirely by control clusters (under rook contiguity), maximizing spatial separation between treatment groups.

Properties: Deterministic, exactly 50% treated, zero spatial correlation in $\mathbf{Z}$.

5.2 Design 2: High Incidence Focus

$$Z_i = \mathbf{1}[X_i > \text{median}(\mathbf{X})]$$

Treatment is assigned to clusters with above-median historical incidence. This represents a strategy that deliberately targets high-burden areas.

Properties: Deterministic given $\mathbf{X}$, exactly 50% treated, treatment correlated with $\mathbf{X}$.

5.3 Design 3: Saturation Quadrants

The grid is partitioned into four spatial quadrants:

$$Q_k, \quad k = 1, 2, 3, 4$$

defined by dividing at the grid midpoint $(G/2, G/2)$. For each randomization:

1. Randomly assign saturation levels $\{0.20, 0.40, 0.60, 0.80\}$ to the four quadrants (one per quadrant, without replacement)
2. Within quadrant Q_k with saturation p_k, randomly select $\lfloor |Q_k| \cdot p_k \rceil$ clusters for treatment

Properties: Stochastic, treatment proportion varies by region, captures varying treatment density.

5.4 Design 4: Isolation Buffer

Treatment is assigned sequentially with spatial constraints:

1. Initialize all clusters as available
2. Randomly select an available cluster and assign treatment ($Z_i = 1$)
3. Remove the selected cluster and all its neighbors from the available set
4. Repeat until no available clusters remain

Properties: Stochastic, treated clusters are never adjacent (under the active contiguity), creates a spatial buffer around each treated unit. Treatment proportion varies across realizations (typically approximately 20–30%).

5.5 Design 5: 2x2 Blocking

The grid is partitioned into non-overlapping 2×2 blocks:

$$B_m = \{s_i : \lfloor (x_i - 1)/2 \rfloor = a_m, \lfloor (y_i - 1)/2 \rfloor = b_m\}, \quad |B_m| = 4$$

Within each block, exactly 2 of 4 clusters are randomly assigned to treatment:

$$\sum_{i \in B_m} Z_i = 2 \quad \text{for all } m$$

Properties: Stochastic, exactly 50% treated, locally balanced (1:1 within each block).

5.6 Design 6: Balanced Quartiles

Clusters are stratified by incidence quartile:

$$S_q = \{s_i : X_i \in [F_X^{-1}((q-1)/4),\ F_X^{-1}(q/4)]\}, \quad q = 1, 2, 3, 4$$

Within each stratum, approximately half the clusters are randomly assigned to treatment:

$$\sum_{i \in S_q} Z_i \approx |S_q|/2$$

Properties: Stochastic, approximately 50% treated, balanced across the incidence distribution. Combines covariate stratification with randomization.

5.7 Design 7: Balanced Halves

Clusters are stratified into two groups by median baseline incidence:

$$S_h = \{s_i : X_i \in H_h\}, \quad h = 1, 2$$

where $H_1 = \{X_i : X_i > \text{median}(X)\}$ (high incidence) and $H_2 = \{X_i : X_i \leq \text{median}(X)\}$ (low incidence).

Within each half, approximately half the clusters are randomly assigned to treatment:

$$\sum_{i \in S_h} Z_i = \lfloor |S_h|/2 \rceil, \quad h = 1, 2$$

Properties: Stochastic assignment. Approximately 50% treated overall. Balanced across the two halves of the incidence distribution. This is a 2-stratum analog of Design 6 (Balanced Quartiles), which uses 4 strata.

5.8 Design 8: Incidence-Guided Saturation Quadrants

The grid is partitioned into four spatial quadrants $Q_1, \ldots, Q_4$ using the same spatial partition as Design 3. Let $\bar{X}_{Q_k} = |Q_k|^{-1} \sum_{i \in Q_k} X_i$ denote the average baseline incidence in quadrant k.

Quadrants are ranked by $\bar{X}_{Q_k}$ in descending order and assigned saturation levels deterministically:

Rank (by $\bar{X}_{Q_k}$)	Saturation p_k
1 (highest)	0.80
2	0.60
3	0.40
4 (lowest)	0.20

Within each quadrant, $\lfloor |Q_k| \cdot p_k \rceil$ clusters are randomly selected for treatment:

$$\sum_{i \in Q_k} Z_i = \lfloor |Q_k| \cdot p_k \rceil$$

Properties: The quadrant-to-saturation mapping is deterministic given incidence; within-quadrant assignment is stochastic. Overall treatment rate approximately 50%. Unlike Design 3, the saturation assignment is guided by incidence data rather than random.

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6 Estimation Methods

6.1 Difference in Means (DIM)

The naive estimator:

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_1 - \bar{Y}_0 = \frac{\sum_{i:Z_i=1} Y_i}{\sum_i Z_i} - \frac{\sum_{i:Z_i=0} Y_i}{\sum_i (1 - Z_i)}$$

Variance estimation (Neyman):

$$\widehat{\text{Var}}(\hat{\tau}_{\text{DIM}}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0}$$

where S_g^2 is the sample variance in group g and $n_g = \sum_i \mathbf{1}[Z_i = g]$.

Confidence interval:

$$\hat{\tau}_{\text{DIM}} \pm z_{0.975} \cdot \sqrt{\widehat{\text{Var}}(\hat{\tau}_{\text{DIM}})}$$

Note: DIM is biased for τ when spillover ($\gamma \neq 0$) or spatial dependence ($\rho \neq 0$) is present, because it ignores the spatial structure of the DGP.

6.2 Maximum Likelihood Estimation (MLE)

The spatial autoregressive model is estimated via maximum likelihood using `spatialreg::lagsarlm()`:

$$Y_i = \rho \sum_j W_{ij} Y_j + \tau Z_i + \gamma \cdot \text{Spill}_i + \beta X_i + \varepsilon_i$$

or in matrix form:

$$\mathbf{Y} = \rho \mathbf{W} \mathbf{Y} + \mathbf{X}_{\text{design}} \boldsymbol{\beta}_{\text{design}} + \boldsymbol{\varepsilon}$$

where $\mathbf{X}_{\text{design}} = [\mathbf{Z},\ \text{Spill},\ \mathbf{X}]$ and $\boldsymbol{\beta}_{\text{design}} = (\tau, \gamma, \beta)^\top$.

The MLE simultaneously estimates ρ and $\boldsymbol{\beta}_{\text{design}}$ via concentrated log-likelihood optimization. Confidence intervals for τ are obtained from the asymptotic covariance matrix:

$$\hat{\tau}_{\text{MLE}} \pm z_{0.975} \cdot \text{SE}(\hat{\tau}_{\text{MLE}})$$

where $\text{SE}(\hat{\tau})$ is extracted from the summary of the fitted `lagsarlm` object.

6.2.1 Oracle vs. Realistic Estimation

A configurable toggle controls whether the true spillover term is included:

- **Oracle** (`include_spill_covariate = TRUE`): The model includes the true Spill($\mathbf{Z}$) covariate. This provides the best-case estimation performance.
- **Realistic** (`include_spill_covariate = FALSE`): The model omits the spillover term, fitting $Y \sim Z + X$ only. This reflects what a researcher would estimate without knowledge of the spillover mechanism.

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7 Simulation Algorithm

7.1 Parameter Space

The simulation sweeps over the following parameter grid:

Parameter	Symbol	Values	Count
Incidence mode	—	iid, spatial, poisson	3 modes
Incidence spatial correlation	ρ_X	0 (iid), 0.20, 0.50	5 configs
Neighbor type	—	rook, queen	2
Spatial autocorrelation	ρ	0.00, 0.01, 0.20, 0.50	4
Spillover magnitude	γ	0.50, 0.60, 0.70, 0.80	4
Spillover type	—	control_only, both	2
Treatment design	—	1, 2, 3, 4, 5, 6, 7, 8	8

Total scenarios: 5 incidence configs $\times$ 2 neighbor types $\times$ 4 ρ $\times$ 4 γ $\times$ 2 spillover types $\times$ 8 designs = **2,560 scenarios**.

7.2 Resample Structure

Each scenario is evaluated over:

- R_d = design resamples (distinct treatment assignment draws)
- R_o = outcome resamples (distinct error realizations per design draw)
- Total iterations per scenario: $K = R_d \times R_o$

Default values:

Estimation Mode	R_d	R_o	K
MLE	25	10	250
DIM	100	1,000	100,000

7.3 Pseudocode

```
BUILD spatial grid -> coords, W_rook, W_queen, listw_rook, listw_queen
SET I_mat = diag(N)

FOR EACH incidence_mode IN {iid, spatial, poisson}:
  SET rho_X_values = {0} if iid, else {0.20, 0.50}

  FOR EACH rho_X IN rho_X_values:
    GENERATE X_matrix (N x R_o) using incidence_mode, rho_X, W_queen
    SET base_incidence = X_matrix[, 1]     # "historical" for design decisions

    FOR EACH nb_type IN {rook, queen}:
      SELECT active W, listw for nb_type

      FOR EACH rho IN {0.00, 0.01, 0.20, 0.50}:
        COMPUTE inv_mat = solve(I_mat - rho * W)
        GENERATE Epsilon_matrix (N x R_o) ~ N(0, sigma^2)
        COMPUTE baseline_part = inv_mat %*% (X_matrix * beta + Epsilon_matrix)

        FOR EACH gamma IN {0.50, 0.60, 0.70, 0.80}:
          FOR EACH spill_type IN {control_only, both}:
            FOR EACH design_id IN {1, 2, 3, 4, 5, 6, 7, 8}:

              SET scenario_seed = hash(incidence_mode, rho_X, nb_type,
                                       rho, gamma, spill_type, design_id)
              set.seed(scenario_seed)

              IF design is deterministic:
                GENERATE Z_matrix with 1 column, replicate to R_d
              ELSE:
                GENERATE Z_matrix (N x R_d)

              INITIALIZE estimates[], ci_lower[], ci_upper[]

              FOR d IN 1:R_d:
                Z = Z_matrix[, d]
                COMPUTE spillover = W %*% Z (apply spill_type mask)
```

```
    COMPUTE trt_effect = inv_mat %*% (tau * Z + spillover)
    Y_sim = trt_effect + baseline_part

    ESTIMATE tau_hat, CI for each of R_o outcome columns

  COMPUTE Bias, SD, MSE, Coverage, Fail_Rate
  STORE results row
```

Critical structural fix: Incidence ($\mathbf{X}$) is generated **once per (incidence_mode, ρ_X) configuration** and held fixed across all spatial parameter combinations (ρ, neighbor type). This reflects the real-world scenario where historical incidence is observed once, and the question is how different spatial models and designs affect estimation quality.

7.4 Deterministic Design Optimization

Designs 1 (Checkerboard) and 2 (High Incidence Focus) produce identical treatment vectors for every resample given fixed $\mathbf{X}$. The algorithm detects this and generates a single assignment, replicating it across R_d to avoid redundant computation.

7.5 Per-Scenario Seeding

Rather than a single global seed, each scenario receives a deterministic seed computed from its parameter combination:

$$\text{seed} = \text{hash}(\text{incidence_mode}, \rho_X, \text{nb_type}, \rho, \gamma, \text{spill_type}, \text{design_id})$$

This ensures reproducibility even if scenarios are added, removed, or reordered.

---

8 Evaluation Metrics

For each scenario, let $\hat{\tau}_1, \ldots, \hat{\tau}_K$ denote the K estimates of τ (excluding convergence failures).

8.1 Bias

$$\text{Bias} = \bar{\hat{\tau}} - \tau, \quad \text{where } \bar{\hat{\tau}} = \frac{1}{K}\sum_{k=1}^{K} \hat{\tau}_k$$

8.2 Standard Deviation

$$\text{SD} = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}\left(\hat{\tau}_k - \bar{\hat{\tau}}\right)^2}$$

8.3 Mean Squared Error

$$\text{MSE} = \frac{1}{K}\sum_{k=1}^{K}(\hat{\tau}_k - \tau)^2 = \text{Bias}^2 + \text{Var}(\hat{\tau})$$

MSE decomposes into squared bias plus variance, capturing both accuracy and precision.

8.4 Coverage Probability

$$\text{Coverage} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{1}\left[\tau \in (\hat{\tau}_{k,L},\ \hat{\tau}_{k,U})\right]$$

where $(\hat{\tau}_{k,L}, \hat{\tau}_{k,U})$ is the 95% confidence interval for the k-th estimate. Nominal coverage is 0.95; under- or over-coverage indicates CI miscalibration.

8.5 Convergence Failure Rate

$$\text{Fail Rate} = 1 - \frac{|\{\text{valid estimates}\}|}{K_{\text{total}}}$$

This captures the proportion of iterations where the MLE optimizer failed to converge (typically due to perfect collinearity between $\mathbf{Z}$ and Spill).

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9 Complete Parameter Table

Parameter	Symbol	Default	Sweep Values	Configurable
Grid dimension	G	10	—	Yes
Number of clusters	N	100	G^2	Derived
True treatment effect	τ	1.0	—	Yes
Incidence coefficient	β	1.0	—	Yes
Error std. deviation	σ	1.0	—	Yes
Spatial autocorrelation	ρ	—	0.00, 0.01, 0.20, 0.50	Yes
Spillover magnitude	γ	—	0.50, 0.60, 0.70, 0.80	Yes
Spillover type	—	—	control_only, both	Yes
Neighbor type	—	—	rook, queen	Yes
Incidence mode	—	—	iid, spatial, poisson	Yes
Incidence spatial corr.	ρ_X	—	0 (iid), 0.20, 0.50	Yes

Parameter	Symbol	Default	Sweep Values	Configurable
Base rate (Poisson)	μ_0	35/100,000	—	Yes
Cluster population	P_0	1,000	—	Yes
Population mode	—	equal	equal, heterogeneous	Yes
Design resamples (MLE)	R_d	25	—	Yes
Outcome resamples (MLE)	R_o	10	—	Yes
Design resamples (DIM)	R_d	100	—	Yes
Outcome resamples (DIM)	R_o	1,000	—	Yes
Design IDs	—	{1,2,3,4,5,6,7,8}	—	Yes
Oracle spillover toggle	—	TRUE	TRUE, FALSE	Yes
Estimation mode	—	MLE	MLE, DIM	Yes

All parameters are configurable to allow investigators to apply these findings to their own study contexts, even when some parameters (e.g., spillover magnitude) would not be known in practice.

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10 References

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